

Onkar Sadekar

■ Postdoctoral Fellow, University of Zürich

Personal Information •

Email: sadekaronkar@gmail.com, onkar.sadekar@uzh.ch

Employment •

2025 - Postdoctoral fellow in the Human Evolutionary Ecology Group lead by Prof. Andrea Migliano, Department of Evolutionary Anthropology, University of Zürich, Zürich, Switzerland.

Education •

2021 - 2025 **Ph.D. in Network Science (*Summa cum Laude*)**: Supervised by Prof. Federico Battiston at the Department of Network and Data Science, Central European University, Vienna, Austria.

Thesis Topic: *Collective dynamics in evolutionary processes and strategic group interactions*

2016 - 2021 **BS-MS Dual Degree**: Indian Institute of Science Education and Research, Pune, India.

Areas of Interest •

Game theory, complex systems, network science, evolutionary anthropology, collective behaviour, cultural evolution

Publications •

10. F. Battiston, V. Capraro, F. Karimi, S. Lehmann, AB. Migliano, **Onkar Sadekar**, A. Sánchez and Matjaž Perc, *Higher-order interactions shape collective human behaviour*. Nat. Hum. Behav. 9, 2441–2457 (2025)
9. **Onkar Sadekar**, A. Civilini, V. Latora, and F. Battiston, *Drivers of cooperation in social dilemmas on higher-order networks*. J. R. Soc. Interface. **22** 20250134 (2025)
8. C. Gunasekaram, F. Battiston, **Onkar Sadekar**, C. Padilla-Iglesias, *et al.*, *Population connectivity shapes the distribution and complexity of chimpanzee cumulative culture*. Science **386**, 6724, 920–925 (2024)
7. **Onkar Sadekar**, A. Civilini, J. Gomez-Gardenes, V. Latora, and F. Battiston, *Evolutionary game selection creates cooperative environments*. Phys. Rev. E **110** 014306 (2024)
6. **Onkar Sadekar**, S. Chowdhary, M.S. Santhanam, and F. Battiston, *Individual and team performance in cricket*. R. Soc. Open Sci. **11** 240809 (2024)
5. A. Civilini, **Onkar Sadekar**, F. Battiston, J. Gomez-Gardenes, and V. Latora, *Explosive cooperation in social dilemmas in higher-order networks*. Phys. Rev. Lett. **132** 167401 (2024)
4. **Onkar Sadekar**, M. Budamagunta, G. J. Sreejith, S. Jain, and M. S. Santhanam, *An infectious diseases hazard map for India based on mobility and transportation networks*, Current Science 121, **9** (2021)

3. S. Saryal, **Onkar Sadekar**, and BK. Agarwalla, *Thermodynamic uncertainty relation for energy transport in a transient regime: A model study*, Phys. Rev. E **103**, 022141 (2021)
2. V. Kumar, **Onkar Sadekar**, and U. Basu, *Active Brownian Motion in two-dimensions under Stochastic Resetting*, Phys. Rev. E **102**, 052129 (2021)
1. **Onkar Sadekar**, and U. Basu, *Zero-current nonequilibrium state in symmetric exclusion process with dichotomous stochastic resetting* J. Stat. Mech. 073209 (2020)

Unpublished works •

2. M. Bozhidarova, Y. Cai, R. Carvalho, D. Cirulli, Q. Dehaene, M. Diaz, A. Krasnokutskaya, B. Pereira, **Onkar Sadekar**, and F. Battiston, *From streaks to synergies: A multi-scale analysis of performance and scoring in the NBA*. arXiv preprint arXiv:2606.27957 (2026).
1. J. Llabrés, **Onkar Sadekar**, F. Malizia, F. Battiston, *Emergence of cooperation in nonlinear higher-order public goods games*. arXiv: 2026:07228

Conferences and Schools •

- IC2S2 Contributed talk at IC2S2-2025, Norrköping.
- NetSci Contributed talk at NetSci-2025, Maastricht.
- NetSciX Contributed talk at NetSciX-2025, Indore.
- VBC Invited talk at the Vienna Behavioural Circle meet 2024, Vienna.
- Como Contributed lightning talk at Lake Como School on Complex Networks-2024, Como.
- NetSciX Contributed talk at NetSciX-2024, Venice.
- NetSci Contributed talk at NetSci-2023, Vienna.
- CompleNet Contributed talk at the CompleNet - 2023, Aveiro.
- CCS Contributed talk at the game theory satellite at the CCS-2022, Palma de Mallorca.

Skills •

- Programming: Comfortable with Python, Julia, C, and Linux-Terminal commands. Well-versed with NetworkX, Pandas, NumPy, and Matplotlib amongst other python libraries.
- Softwares: Proficient in Inkscape, gnuplot, and $\text{\LaTeX}$.

Scholarships & Awards •

- 2026 CEU Award for Best dissertation.
- 2025 CEU Award for Advanced Doctoral Students to “recognize exceptional performance and acknowledge outstanding research achievement”.
- 2021 - 2025 Full scholarship and tuition fee waiver at Central European University to pursue Ph.D. in Network and Data science.
- 2017 Gold Medal in National Graduate Physics Exam (NGPE)-2017, conducted by Indian Association of Physics Teachers (IAPT). Placed among top 5 out of 15,000 students.
- 2016-2021 INSPIRE scholarship by Government of India for pursuing a Bachelor and Masters in Science.

Teaching •

- Fall 2022 TA for *'Introduction to complex systems'* course for Bachelors students.
- Fall 2023 TA for *'Introduction to complex systems'* course for Bachelors students.
- Winter 2024 TA for *'Introduction to quantitative social science'* course for Bachelors students.
- Fall 2024 TA for *'Introduction to complex systems'* course for Bachelors students.
- Fall 2024 TA for *'Research seminar'* course for Masters students.
- Fall 2025 TA for *'Current debates in evolutionary biology and anthropology'* course for Bachelors and Masters students.

Service to community •

- yrCSS **Advisory board member** for young researchers of the complex systems society (2026-2028). [Website](#)
- Server admin **Server admin** from September 2022 to September 2023 for the Department of Network and Data Science Computing Cluster at Central European University.
- NetSci-2023 **Web and Social Media head** for NetSci-2023, the international conference on Network science. <https://netsci2023.wixsite.com/netsci2023>